H

Year 6 Preliminary Examination
Higher 2

BIOLOGY

9648/01

Paper 1 Multiple Choice

27th September 2016

1 hour 15 min

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name and shade your Index Number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **forty** questions in this paper. Answer all questions. For each question there are four possible answers **A, B, C**, and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

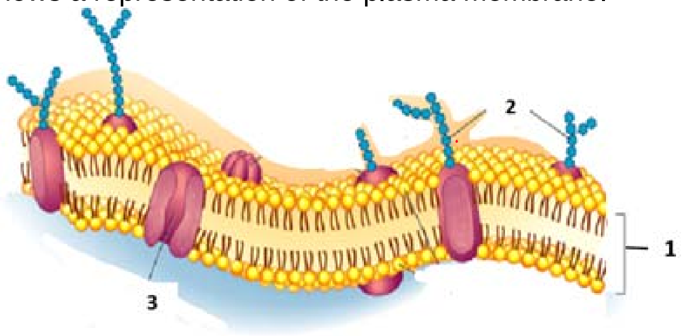
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

(Erase all mistakes completely. Do not bend or fold the OMR Answer Sheet).

This document consists of **29** printed pages.

Raffles Institution
Internal Examination

1. The diagram below shows a representation of the plasma membrane.



Which of the following structures are correctly matched with their role(s)?

	cell-cell recognition	maintenance of resting membrane potential	uptake of steroid hormones
A	**2** only	**1** and **3** only	**1** only
B	**2** only	**1** only	**1** and **3** only
C	**2** and **3** only	**1** only	**3** only
D	**3** only	**1** and **3** only	**1** only

2. The figure below shows an enzyme-catalysed pathway.

Oxaloacetic acid
CO_2 ↙ Enzyme 1
↓
Succinic acid
2NADH ↙ Enzyme 2
↓
Fumaric acid
↓ Enzyme 3
Malic acid

The addition of malonic acid results in no change in the concentration of oxaloacetic acid, an accumulation of succinic acid, and a very low concentration of both fumaric acid and malic acid.

What does this information indicate about malonic acid?

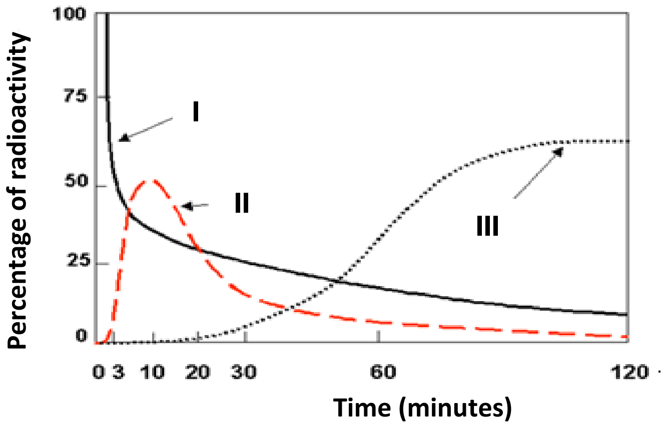
A It is an inhibitor of enzyme 1.

B It catalyses the formation of succinic acid.

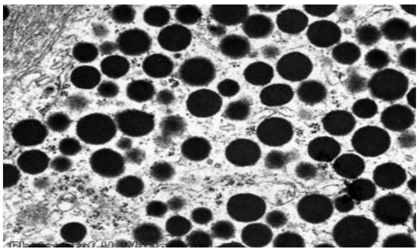
C It is an inhibitor to enzyme 2.

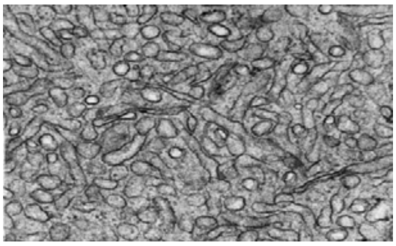
D Malonic acid is reduced in the process.

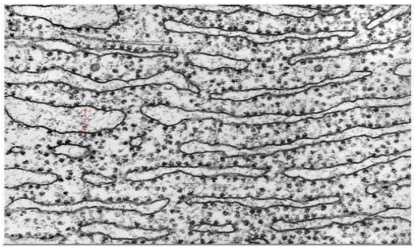
3. A pulse-chase experiment was used to trace the path taken by amino acids in a cell over a period of time. During the experiment, a cell producing salivary amylase was grown in a culture containing radioactive amino acids for a few minutes. The percentage of radioactivity at the various organelles was then measured and the results of the experiment are shown in the graph below.

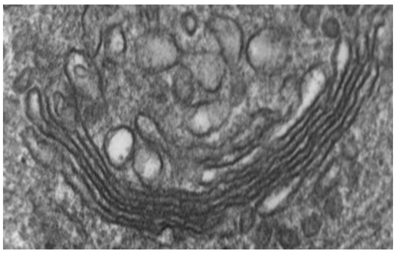


The micrographs of four different organelles from the cell producing salivary amylase are shown below.


P


Q


R

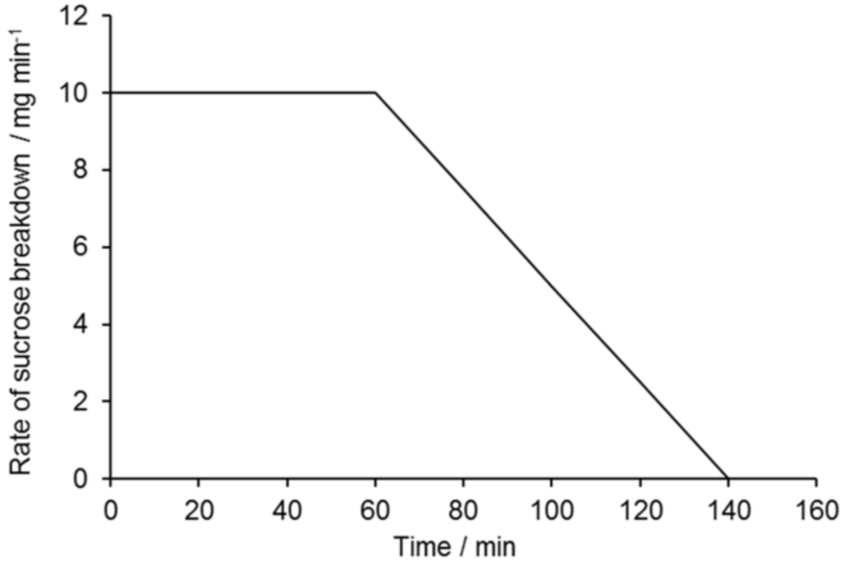

S

Each graph corresponds to an organelle in the cell where the radioactivity was measured.

Which of the organelles shown above are correctly matched with the graphs?

	graph I	graph II	graph III
A	R	S	P
B	P	R	S
C	Q	R	S
D	R	S	Q

4. The graph shows the results of an investigation using invertase, an enzyme that breaks down sucrose into glucose and fructose. 1 g of sucrose was dissolved in 100 cm^3 of water and 2 cm^3 of a 1% invertase solution was added.



Which conclusion can be drawn from this information?

A Between 0 and 60 min, the concentration of the substrate remains constant.

B After 60 min, the concentration of enzymes becomes the limiting factor.

C At 140 min, some of the enzyme molecules are denatured.

D Between 60 and 140 min, the concentration of the substrate is the limiting factor.

5. Vitamin C adds hydroxyl groups to two amino acids, proline and lysine. Without the presence of Vitamin C, the production of collagen is disrupted.

This is due to the inability to form

A the tertiary structure of tropocollagen.

B collagen fibrils.

C the secondary structure of collagen.

D disulfide bonds.

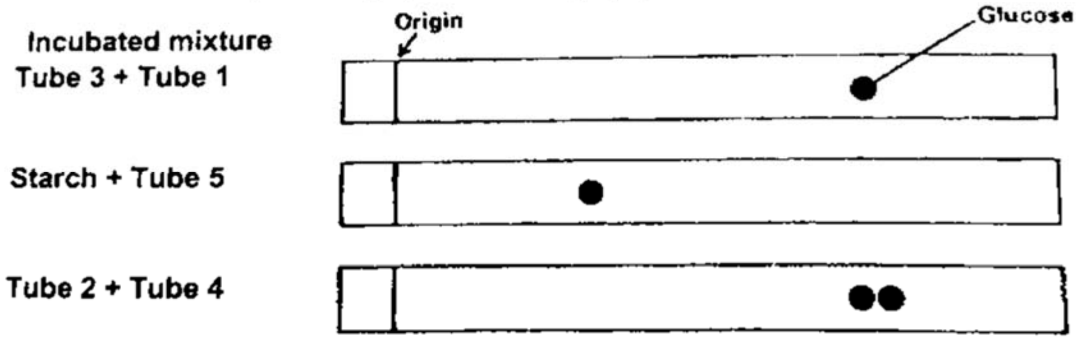
6. 3 samples of common carbohydrates and 3 samples of enzymes were randomly mixed in different combinations in 5 different tubes. The following statements are some observations of various tests that were conducted on the contents of the 5 tubes.

I Sample in tube 2 is soluble in water. Sample in tube 3 was insoluble in water.

II Tube 5 tested positive with Biuret's test.

III All 6 individual samples tested negative with Benedict's test. However, certain mixtures showed positive test after incubation with other tubes.

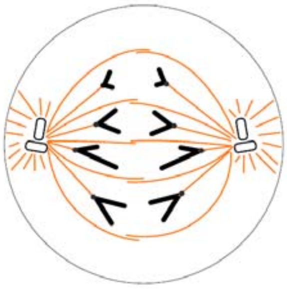
IV The mixtures were subjected to paper chromatography, and the results were shown below.



Which of the following correctly shows the contents of each tube?

	tube 1	tube 2	tube 3	tube 4	tube 5
A	amylase	cellulose	sucrose	cellulase	sucrose
B	cellulase	sucrose	cellulose	sucrase	amylase
C	cellulose	sucrose	cellulose	sucrase	amylase
D	sucrose	cellulose	sucrase	amylase	cellulase

7. The following diagram shows a stage during cell division in a eukaryotic diploid cell.



Which of the following statement(s) is/are false?

I The diagram represents a stage in mitosis.

II The diagram represents a stage in meiosis **II**.

III The chromosome number of the cell at interphase is 2n = 4.

IV This process occurs in the root tips of plants.

A **II** and **III** only

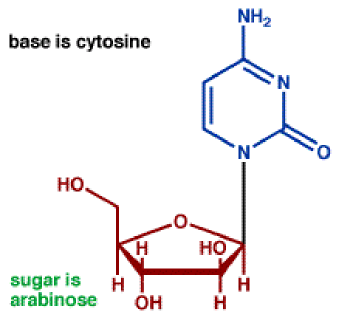
B **I**, **III** and **IV** only

C **I** and **IV** only

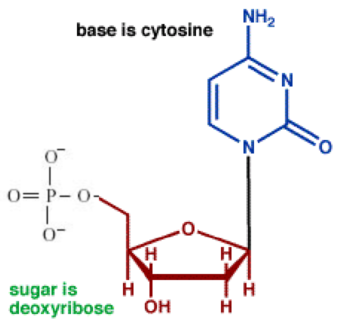
D **III** only

8. Cytarabine is a drug used to treat certain cancers. It consists of a cytosine base and an arabinose sugar.

The diagram below shows the structures of cytarabine and the deoxyribonucleotide, deoxycytidine triphosphate (dCTP).



Cytarabine



Deoxycytidine triphosphate

Which of the following statements are true?

I Cytabarine has a hydroxyl group attached to carbon number 4 of its sugar while deoxycytidine triphosphate has a phosphate group attached to carbon number 4 of its sugar.

II Cytarabine prevents DNA replication.

III Deoxycytidine triphosphate molecule has a free 3' hydroxyl group that can form a phosphodiester bond with another ribonucleotide during DNA replication.

IV Cytarabine has a greater effect on cancer cells than healthy cells as cancer cells divide faster than healthy cells.

A **I** and **II** only

B **II** and **IV** only

C **I** and **III** only

D **II, III** and **IV** only

9. Fig. 9.1 and Fig. 9.2 are electron micrographs that show ribosomes (dark circular structures) involved in protein synthesis. One figure illustrates protein synthesis in a eukaryotic cell while the other illustrates protein synthesis in a prokaryotic cell.

Fig. 9.1

Fig. 9.2

Which of the following statement(s) is/are false?

I Fig. 9.1 illustrates a process that occurs in prokaryotic cells while fig. 9.2 illustrates a process that occurs in eukaryotic cells.

II The arrows in both fig. 9.1 and fig. 9.2 are pointing to the chromosomal DNA.

III Complementary base pairing occurs between the rRNA in the mRNA binding site of the small ribosomal subunit and the mRNA.

IV The rRNA molecule in the ribosomal subunit catalyses the formation of a peptide bond between the amino group of the incoming amino acid at the P site and the carboxyl end of the growing polypeptide chain at the A site.

A **II** only

B **II** and **IV** only

C **I** and **III** only

D All of the above

10. Sickle cell anaemia is caused by a mutation in the gene that codes for the β-globin polypeptide of haemoglobin.

The sequence of bases below is a small section of the template strand of DNA for both the normal allele (HbA) and the sickle cell allele (HbS).

HbA allele CTGACTCCTGAGGAGAAGTCT

HbS allele CTGACTCCTGTGGAGAAGTCT

How will the mutation in the HbS allele result in the production of a non-functional β-globin polypeptide?

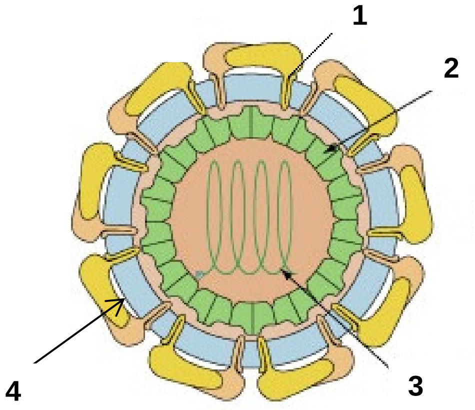
A The mRNA transcribed from the HbS allele will contain the codon CAC instead of the codon CTC.

B All the amino acids coded for after the mutation will differ from those in the HbA protein.

C A tRNA molecule with the anticodon GUG will hydrogen bond to the altered codon on mRNA.

D The ribosome will be unable to continue translation of the HbS mRNA after the altered codon.

11. The Zika virus is a type of flavivirus. Its replication cycle is similar to that of the influenza virus. The diagram below shows the structure of the Zika virus.



Which of the following correctly matches the numbered structures?

	1	2	3	4
A	transmembrane protein	capsid	DNA genome	lipid bilayer
B	viral glycoprotein	capsomere	RNA genome	viral envelope
C	receptor	matrix protein	viral genome	capsid
D	viral glycoprotein	capsomere	nucleoprotein	matrix protein

12. During the replication cycle of the human immunodeficiency virus (HIV), the polyprotein that is produced is cleaved by a viral protease enzyme, producing several smaller peptides. This viral enzyme is the target of anti-HIV drugs.

Which feature is essential for the success of these drugs?

A A complex structure that inhibits many types of enzymes.

B A molecule containing a heavy metal atom that is non-competitive inhibitor of enzymes.

C A protein that can act as a competitive inhibitor of protease enzymes.

D A specific structure that inhibits only viral protease.

13. Which of the following statement(s) concerning *trp* operon is/are true?

I A deletion mutation of the operator will lead to the constitutive production of tryptophan.

II There is one start and one stop codon in the mRNA of *trp* operon.

III The repressor is inactive in the presence of excess tryptophan.

IV The mRNA codes for 3 polypeptides involved in the synthesis of tryptophan.

A **I** only

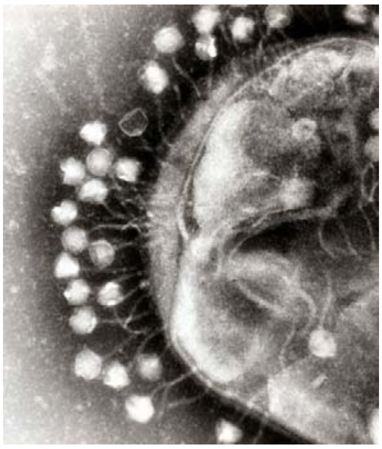
B **I, II** and **III** only

C **II** and **III** only

D **I** and **IV** only

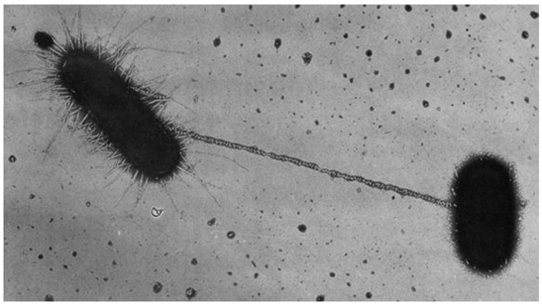
14. The following electron micrographs show processes involving bacteria.

X



Y

Z



Which of the following statement(s) is/are true?

I All the processes, **X, Y** and **Z** increase the genetic diversity in bacteria.

II The rolling circle mechanism of DNA replication occurs in processes **Y** and **Z**.

III Process **X** may lead to the introduction of new bacterial DNA into the bacteria.

IV DNA replication occurs in processes **Y** and **Z**.

A **I**, **II** and **III** only

B **II** and **III** only

C **I** and **IV** only

D **III** and **IV** only

15. The following processes are different means by which gene expression can be regulated.

I amplification of a specific gene by rolling circle replication

II small effector molecules bound to activator protein controlling an inducible operon

III regulatory protein bound to control element causing spacer DNA to bend such that this regulatory protein interacts with TATA box directly

IV removal of acetyl groups from lysine residue of histones

Which of the above processes would result in the upregulation of gene expression?

A **I** and **II** only

B **II** and **IV** only

C **I**, **II** and **III** only

D All of the above

16. *BRCA1* and *BRCA2* genes codes for gene products with DNA repair functions. Mutations in these two genes increase the risk of female breast cancer or ovarian cancer. It is observed that individuals who inherit mutations in *BRCA1* and *BRCA2* genes tend to develop breast or ovarian cancer at a younger age.

Which of the following statements provides the best explanation for such observation?

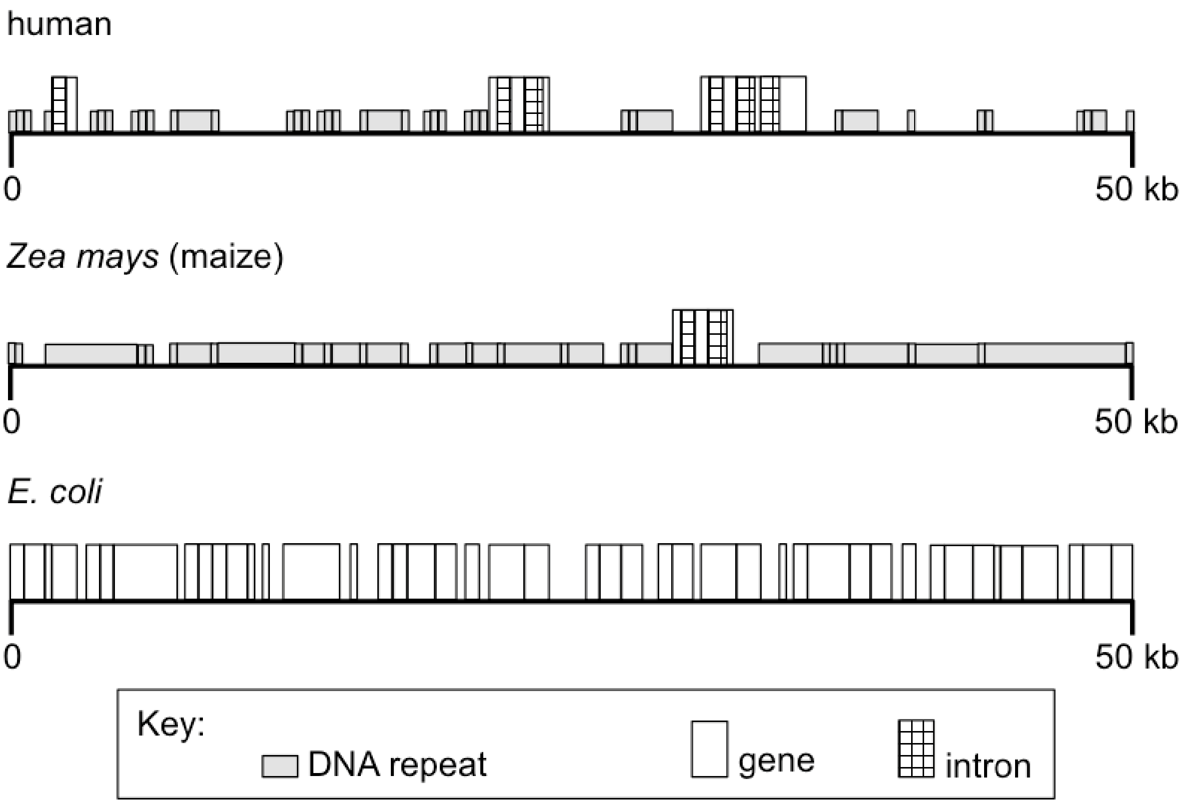
A Both *BRCA1* and *BRCA2* genes are proto-oncogenes. When mutated, their non-functional protein products cannot repair damaged DNA.

B Other oncogenes may accumulate mutations quickly to form proto-oncogenes as there are no protein products to carry out DNA repair.

C Development of cancer is a multistep process that requires accumulation of mutations in more than 1 gene.

D Individuals have mutations in a copy of each tumour suppressor gene, *BRCA1* and *BRCA2*. Thus less time is required for a single cell to acquire another loss-of-function mutation in the other copy of the tumour suppressor gene and a gain-in-function mutation in a proto-oncogene to become cancerous.

17. The diagram shows 50 kb segments of the human, *E.coli* and *Zea mays* genomes.



Which one of the following statements can be concluded from the above results showing the genetic organisation of a 50 kb portion of the human, *E.coli* and *Zea mays* genomes?

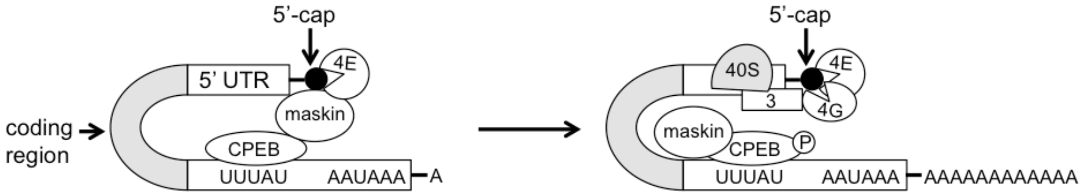
A More complex organisms have lower gene density.

B Organisms with smaller chromosome number have higher gene density.

C *Zea mays* has a higher density of DNA repeats as compared to humans and *E. coli*.

D The presence of introns in DNA of eukaryotes allows alternative splicing to occur to synthesise as many proteins as prokaryotes.

18. During development, an immature egg cell of *Xenopus* has many translationally dormant mRNAs in its cytoplasm. The figure below shows how one of these mRNAs becomes active upon maturation of the egg cell. In the figure, 4E, 4G and 3 are eukaryotic translation initiation factors.



translationally dormant mRNA in immature egg cell

translationally active mRNA in mature egg cell

Which one of the statements describes how translation is repressed in the immature egg cell?

A Mutation in the 5' UTR of mRNA in immature egg cell causes maskin protein to temporary interact with the 5' cap until its original mRNA sequence is restored in mature egg cell.

B CPEB needs to be phosphorylated in order for maskin protein to detach from it and enable large ribosomal subunits and eukaryotic translation initiation factors to bind to 5' of mRNA.

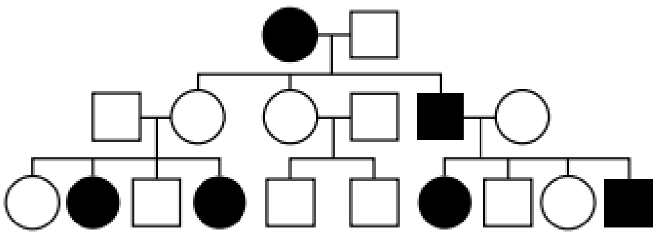
C Maskin protein interacts with the 5' cap of mRNA in immature egg cell to prevent formation of translation initiation complex that is made up of small ribosomal subunit and a set of eukaryotic translation initiation factors.

D Absence of poly(A) tail of mRNA in immature egg cell prevents assembly of translation initiation complex.

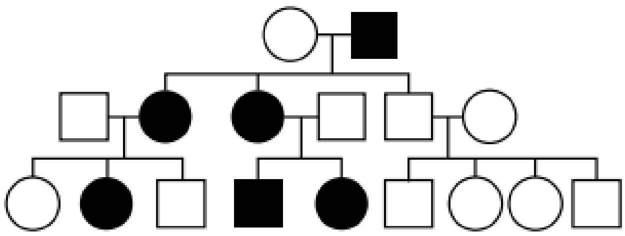
19. Kearns-Sayre syndrome is a rare genetic trait caused by a deletion of up to 10 000 nucleotides from the mitochondrial DNA (mtDNA). Most individuals with this syndrome have weak eye muscles, drooping eyelids, vision loss and, often, short stature.

Which pedigree shows a family affected by Kearns-Sayre syndrome?

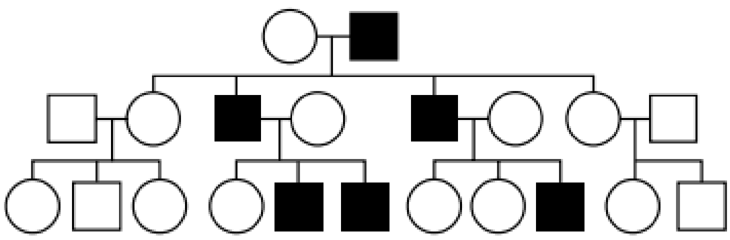
A



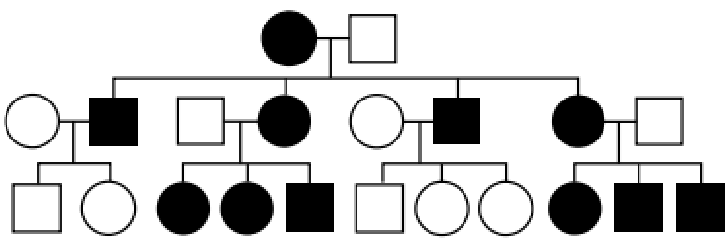
B



C



D



20. Flamingos are birds that live by lakes. The feather colour of flamingos may vary from white to pink to red. To investigate the inheritance of feather colour, a scientist performed the following crosses and recorded the feather colour of all the offspring when they were one year old. The diet of the offspring was also recorded.

cross	**feather colour of parents**	**feather colour of all one-year old offspring**	**diet of offspring**
1	white x white	white	aquatic plants
2	red x white	white	aquatic plants
3	white x white	pink	algae and crustaceans
4	red x white	pink	algae and crustaceans

Based on the above information, which of the following is a correct conclusion?

A Both the parents in cross 1 must be homozygous for white feather colour.

B White feather colour is recessive to red feather colour.

C The feather colour of flamingos is influenced by their environment.

D Two pink-feathered parents would only produce one year old offspring with pink feathers.

21. In guinea pigs, the genes for hair length and hair type have the following alleles:

allele S : long hair
allele s : short hair
allele W : straight hair
allele w : wavy hair

A breeder first carried out a cross between pure-breeding long, straight hair guinea pigs and short, wavy hair guinea pigs. The offspring of this cross were then subjected to many test crosses to determine if the two genes were linked.

If the two genes were closely linked, which of the following is a likely result of the test crosses?

A There will be more guinea pigs with long, wavy hair than short, wavy hair.

B There will be more guinea pigs with short, straight hair than short, wavy hair.

C There will be approximately equal numbers of long, straight hair and long, wavy hair guinea pigs.

D There will be approximately equal numbers of long, straight hair and short, wavy hair guinea pigs.

22. In an experiment, chloroplast extracts were first treated with a chemical that 'snatches' away the electron that was accepted by the electron acceptor in photosystem I. The extracts were then treated with 2 hours of light and were provided with ample carbon dioxide and water.

Which of the following correctly shows the products that were formed after the experiment?

	O_2	ATP	reduced NADP	glucose
A	+	+	–	–
B	–	+	+	+
C	+	–	–	–
D	–	–	+	–

Key: (+) = present, (–) = absent

23. The blue dye DCPIP can be converted to colourless DCPIP as shown below:

DCPIP (blue) → reduced DCPIP (colourless)

A suspension of chloroplasts was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were then prepared and treated in the following ways.

tube	contents	treatment	colour	
			at start	after 20 minutes
1	1 cm^3 chloroplast suspension + 5 cm^3 DCPIP	illuminated strongly	blue green	green
2	1 cm^3 buffer solution + 5 cm^3 DCPIP	illuminated strongly	blue	blue
3	1 cm^3 chloroplast suspension + 5 cm^3 DCPIP	left in the dark	blue green	blue green

Which one of the following statements is a possible conclusion for the observation above?

A Electron transfer from reduced NAD to DCPIP causes the decolourisation of DCPIP.

B NADP was oxidised and the electron was used to decolourise DCPIP.

C Light dependent reaction which occurs in the chloroplasts yield free electrons which reduced DCPIP.

D Either strong illumination or the buffer solution used in the extraction of chloroplasts could oxidise DCPIP.

24. Six tubes were set up as shown in the table and incubated.

tube	contents
1	glucose + homogenized plant cells
2	glucose + mitochondria
3	glucose + cytoplasm from liver cells lacking organelles
4	pyruvate + homogenized liver cells
5	pyruvate + mitochondria
6	pyruvate + cytoplasm from liver cells lacking organelles

Which of the following tubes will contain lactate?

A **1** and **3** only

B **2, 3, 5** and **6** only

C **4, 5,** and **6** only

D **3** and **6** only

25. One form of fur color in mice is controlled by the interaction of two gene resulting in three phenotypes: agouti (alternating dark and light bands), black, and albino. Both genes affect the same trait (fur color). One gene controls the formation of pigment (A) and the other controls the distribution of the pigment (B) when it is produced.

Two agouti mice which bred repeatedly produced the following offspring: 46 agouti, 16 black, and 23 albino.

Which of the following statements are true?

I The gene interaction pattern is indicative of recessive epistasis.

II The gene interaction pattern is indicative of dominant epistasis.

III The genotype of both the agouti parents is AaBb.

IV The offspring genotypic ratio is: 9A_B_: 3A_bb: 3aaB_: 1aabb
The offspring phenotypic ratio is: 9 agouti: 3 black: 4 albino

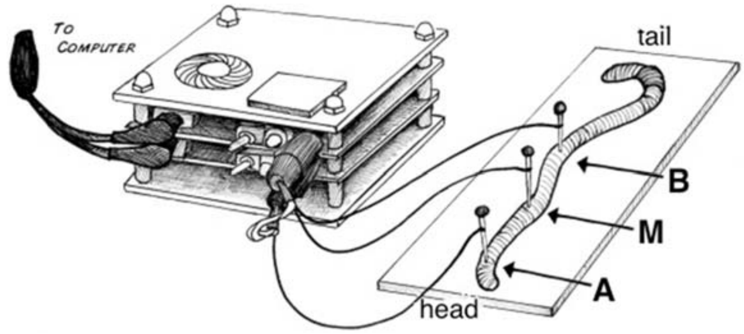
A **I** and **III** only

B **II** and **III** only

C **II** and **IV** only

D **I**, **III** and **IV** only

26. An experiment that records the change in the membrane potential from the giant fibre of an anesthetised earthworm was performed. As shown in the simplified diagram below, an electrical stimulus was applied to point M (middle of the axon) while recording electrodes were present at regions A and B.

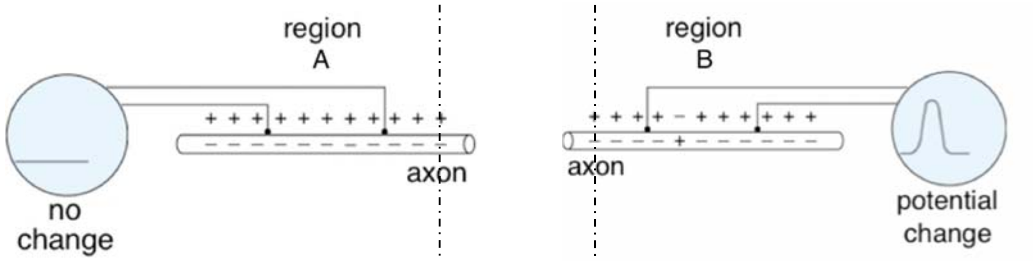


The following shows the possible results at the recording electrodes at regions A and B after a stimulus was applied to point M.

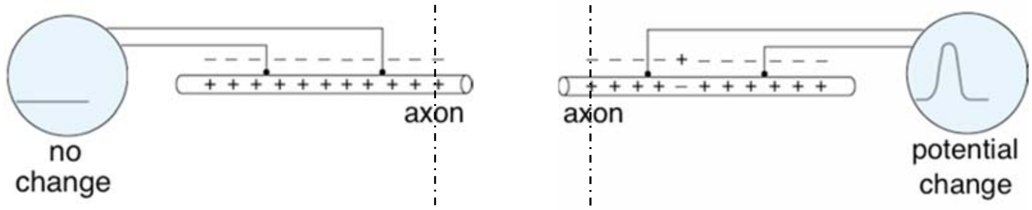
Which of the following is correct?

	relative charges across membrane at region A	**M**	**relative charges across membrane at region B**

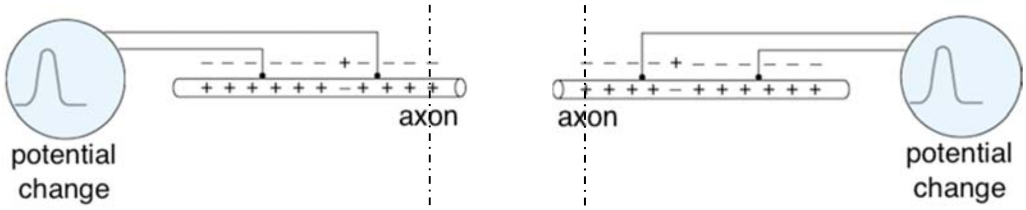
A



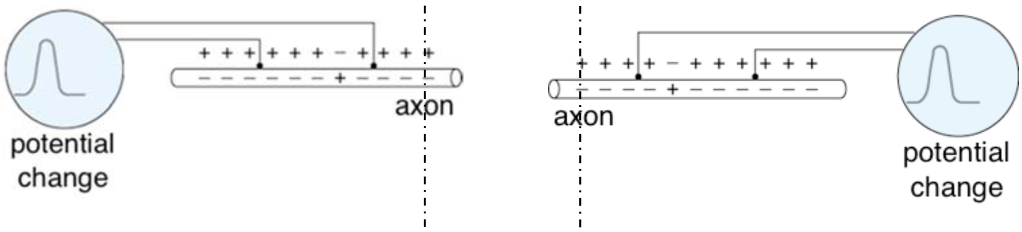
B



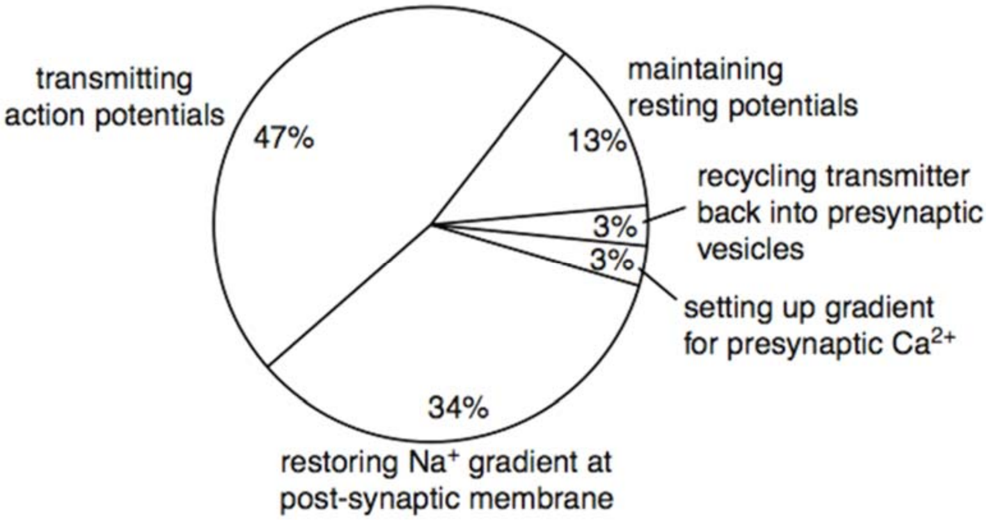
C



D



27. The pie chart shows the percentage of energy used by a myelinated motor neurone for various processes related to impulse transmission.



Which of the following statements about the neurone can be inferred from the results shown in the pie chart?

A The percentage of energy required at the nodes of Ranvier is 60%.

B 3 sodium ions move out to extracellular and 2 potassium ions move in to cytosol of axon to maintain resting potentials.

C 34% of energy is used to transport Na^+ into the post-synaptic neurone to restore the Na^+ gradient.

D 3% energy is required to open the voltage-gated calcium ion channel to enable influx of calcium ions into the presynaptic knob.

28. All of the following statements about events involved in glycogen metabolism are true except

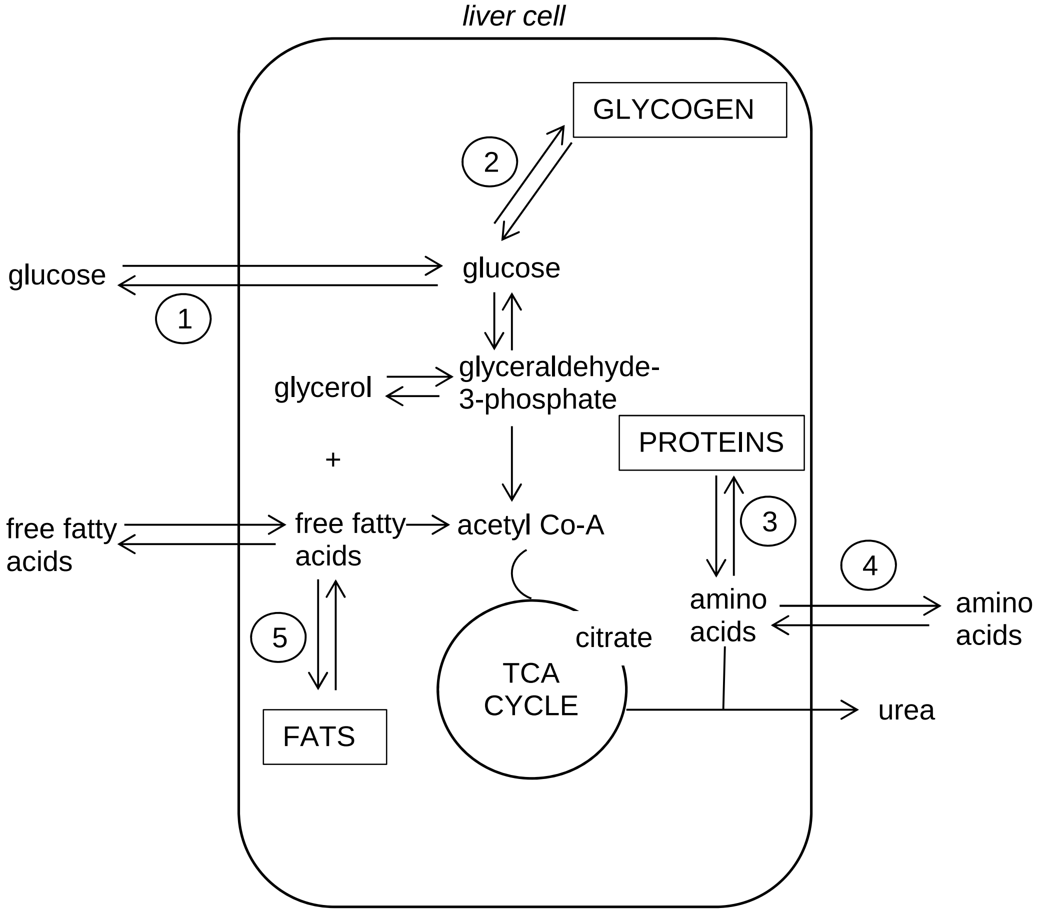
A cAMP-activated protein kinase stimulates glycogen synthesis.

B Kinases are activated by phosphorylation.

C cAMP levels are raised during glycogen hydrolysis.

D Cross-phosphorylation of receptor subunits leads to glycogenesis.

29. The diagram below shows the biochemical pathways in a liver cell. Some of the points where hormones affect the pathways are labelled 1 to 5.



At which numbered points would the hormone insulin accelerate the pathways in the directions indicated?

A 1, 2 and 3 only

B 1, 2 and 5 only

C 2, 3 and 4 only

D 2, 3 and 5 only

30. Which of the following statement(s) about receptor tyrosine kinases (RTKs) is/are true?

I The insulin receptors has a 7-pass transmembrane domain.

II RTKs can auto-phosphorylate tyrosine residues on other proteins.

III RTKs have transmembrane domains.

IV RTKs are subunits that can dimerise.

A **III** and **IV** only

B **II** and **IV** only

C **I, III** and **IV** only

D All of the above

31. Two areas of molecular biology that have received considerable attention in evolutionary studies are the genetic code and cytochrome C. Cytochrome C is an essential component of all respiratory electron transport chains.

Which statements lend evidence to the following 2 ideas?
Idea 1: All living organisms are related, and
Idea 2: There is a single, rather than a multiple, origin of life?

1	The almost universal nature of the genetic code is a result of evolutionary convergence from multiple lineages.
2	The sequence of amino acids in cytochrome C is similar in organisms that are from similar environments or with similar metabolic demands.
3	The majority of organisms have the same, or similar, amino acid sequences for cytochrome C.
4	When transferred into a very dissimilar organism, a gene coding for cytochrome C will lead to the expression of a protein that will function in the other organism.

A **1** and **2** only

B **2** and **3** only

C **3** and **4** only

D **1**, **3** and **4** only

32. Which of the following shows the correct sequence of events?

I	adaptation of a population	➔	competition and predation leading to natural selection	➔	behavioural isolation	➔	allopatric speciation
II	adaptation of a population	➔	competition and predation leading to natural selection	➔	physiological isolation	➔	allopatric speciation
III	competition and predation leading to natural selection	➔	physiological isolation	➔	adaptation of isolated populations	➔	sympatric speciation
IV	competition and predation leading to natural selection	➔	geographical isolation	➔	adaptation of isolated populations	➔	allopatric speciation

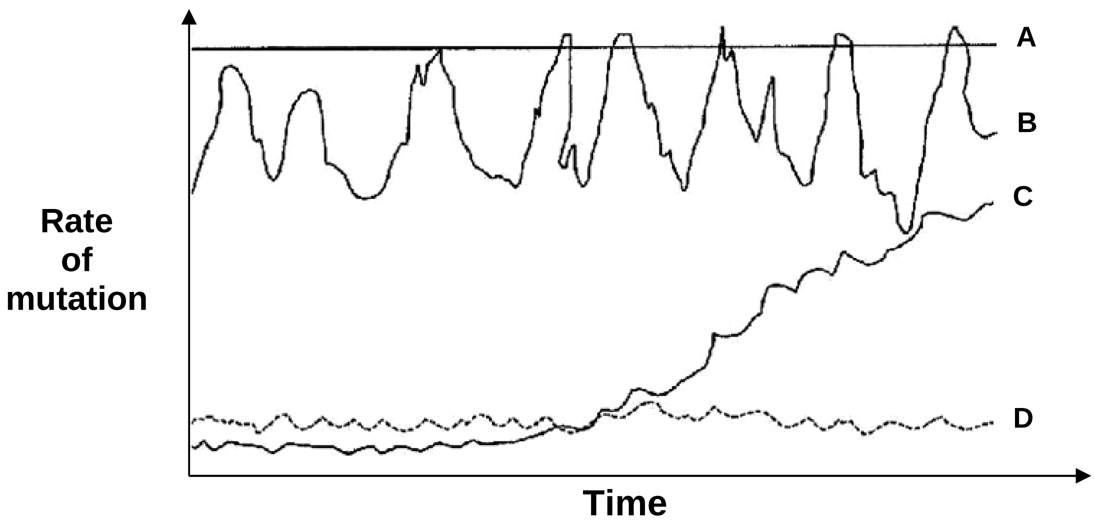
A **III** only

B **I** and **II** only

C **III** and **IV** only

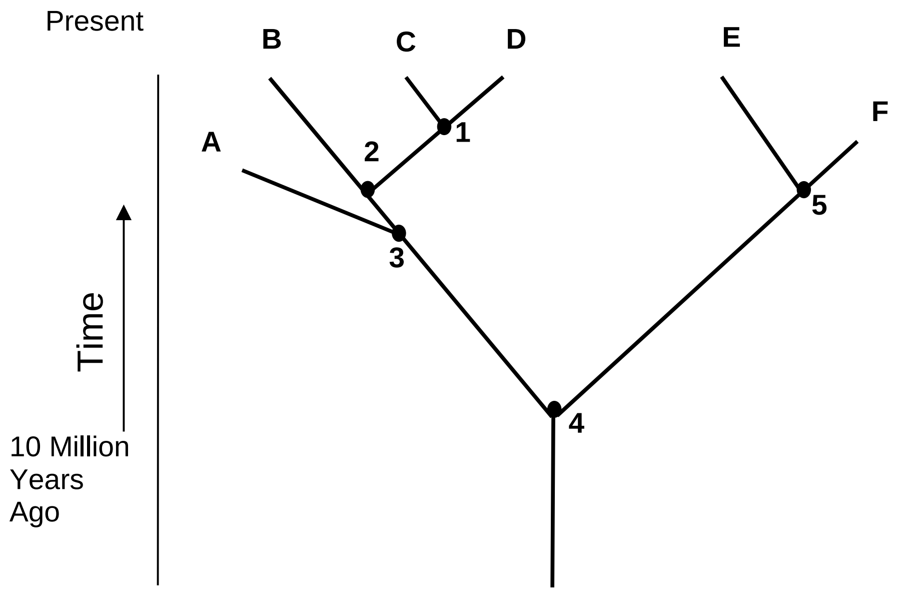
D **I**, **III** and **IV** only

33. The rate of mutation of 4 different genes were investigated and the results shown below.



Which pattern of mutation is most suitable if one desires to use a gene as a molecular clock to determine evolutionary relatedness of different species?

34. The diagram below shows a phylogenetic tree.



Which of the following statements are true?

I Organisms A, B, C, D, E and F are of different species, derived from a common ancestor.

II Organisms A and B are more closely related than organisms E and F.

III Organisms A and F are extinct.

IV Organisms C share more homologous structures with B than with D.

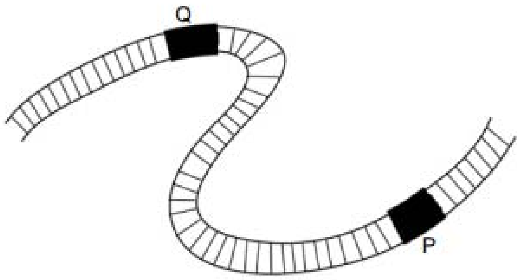
A **I**, **II** and **III** only

B **I** and **III** only

C **II** and **IV** only

D **III** and **IV** only

35. The genome of a small virus is depicted below, showing the positions of restriction sites P and Q for two different restriction enzymes.



The length of DNA fragments obtained when using these restriction enzymes is shown in the table alone.

restriction site	**restriction enzyme**	**length of DNA fragments obtained (kb)**
Q	*Eco*RI	3, 7
P	*Bam*HI	8, 2

If both *Eco*RI and *Bam*HI are used to cut this viral DNA, what will the length (in kb) of the DNA fragments obtained be?

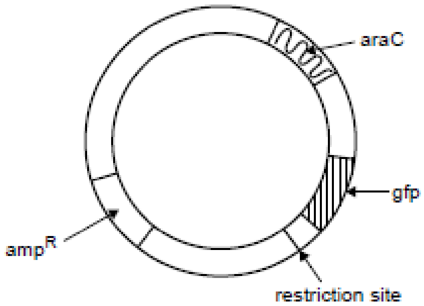
A 1, 2, 7

B 1, 3, 6

C 2, 3, 5

D 2, 3, 7

36. A diagram of a plasmid used in cloning is shown below.

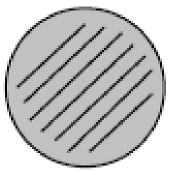
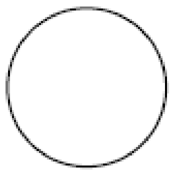
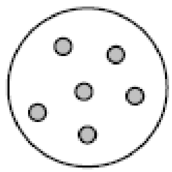
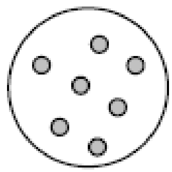


This plasmid contains a restriction site and three genes:
- *amp*^{R} – confers resistance to the ampicillin antibiotic
- *gfp* – encodes the green fluorescent protein (GFP), which fluoresces under UV light
- *araC* – encodes a protein that enables the expression of gfp when arabinose is present

E.coli were transformed with the above plasmids.

Untransformed bacteria were grown on nutrient agar plates W and X, while transformed bacteria were grown on nutrient agar plates Y and Z.

The results of the experiment are shown in the table below.

Plate	W untransformed bacteria only	X untransformed bacteria only	Y transformed bacteria	Z transformed bacteria
Diagram of plate				
Added to plate	nutrient agar only	nutrient agar and ampicillin	nutrient agar, ampicillin and arabinose	nutrient agar and ampicillin
Description of result	lawn of bacteria	no growth	bacterial colonies present	bacterial colonies present

Which one of the following statement(s) is/are true?

I Plate W shows that the nutrient agar promoted the growth of viable bacteria.

II Plate Y and Z contained bacteria that are ampicillin resistant and are able to produce arabinose.

III Plate Y contains bacteria that fluoresce under UV light.

IV Plate X is the negative control.

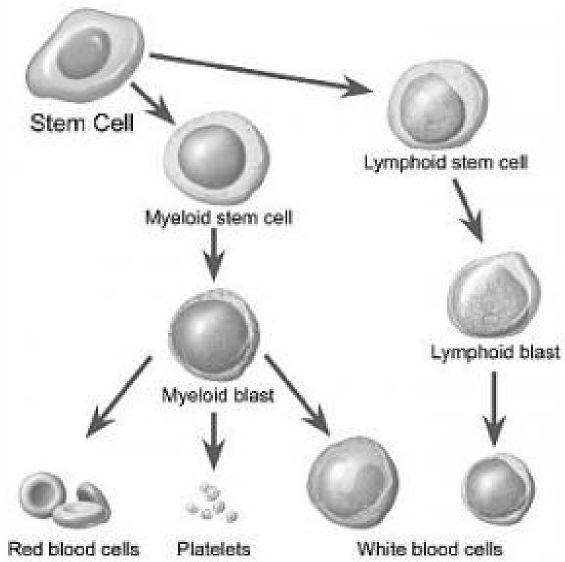
A **III** and **IV** only

B **III** only

C **I**, **III** and **IV** only

D **I** and **III** only

37. The following diagram shows how a stem cell can differentiate into different specialized cell types.



Which of these statements is false with regards to the stem cells shown?

A The stem cells are multipotent.

B The stem cells can be found in both a developing fetus and adult.

C The stem cells can differentiate into the endothelial layers in the blood vessels in the adult body.

D The stem cells may be used in to treat a patient suffering from SCID.

38. Which statement(s) explain(s) the limited success of somatic gene therapy as a permanent cure for genetic disorders in human populations?

I Somatic cells cannot pass the modified gene on to any offspring.

II The treatment does not last long as the treated somatic cells die and are replaced.

III Post-translational modification may be missing in the treated cells.

A **I** only

B **I** and **II** only

C **II** and **III** only

D All of the above

39. Corn is a major crop grown in Europe. In the past, it was either ruined by attack from the corn borer larvae or intensively sprayed with pesticides each year. The biotech company Novartis gained approval to insert a *Bt* gene into corn. This gene codes for a protein that kills the larvae feeding on the corn. The genetically engineered *Bt* corn initially thrived without the addition of any pesticides, but later suffered damage from pests again.

Which of the following are possible reasons why the yield of *Bt* corn decreased again over a few generations?

I A strain of resistant larvae has emerged due to a chance mutation. The frequency of the gene conferring resistance as well as the number of larvae with resistance has increased. The *Bt* gene is now ineffective.

II Corn is being attacked by other pests which are not killed by the *Bt* gene protein.

III Mutations in corn may have led to the loss of the *Bt* allele.

A **I** and **II** only

B **I** and **III** only

C **II** and **III** only

D All of the above

40. Which of the following matches between the process and its outcome is incorrect?

	process	**outcome**
A	anther culture	production of homozygous plants
B	embryo culture	production of distantly related plants
C	protoplast culture	formation of hybrid plants from parent plants of different species
D	callus culture	generation of differentiated tissues

End of Paper